

Vol 13 Chapter 6 — Biochemistry Bridge

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Biochemistry Bridge

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

Biochemistry — the chemistry of living systems including enzyme catalysis, metabolic pathways, ATP energy currency, photosynthesis, respiration — is the substrate K-type rearrangements of biological macromolecules under living-system boundary conditions, bridging Vol 12 chemistry to Vol 13 biology.

Level 2 — graduate-physicist precision

6.1 Enzyme catalysis

Enzymes lower activation energy by factors of 10^5 - 10^{17} (Vol 12 Ch 5).

Michaelis-Menten kinetics:

$$v = \frac{V_{\max}[S]}{K_M + [S]}$$

with $V_{\max}$ maximum rate, K_M Michaelis constant.

Active site provides substrate-K-type-optimal saddle path (Vol 12 Ch 5 transition state).

6.2 ATP energy currency

Adenosine triphosphate. Hydrolysis: $\text{ATP} \rightarrow \text{ADP} + \text{Pi} + 30.5 \text{ kJ/mol}$ (at standard conditions; $\sim 50 \text{ kJ/mol}$ in vivo).

Universal energy currency across all known life. $\sim 10^{21}$ ATP molecules / human / day.

Synthesized by ATP synthase (mitochondria) via chemiosmotic proton gradient (Mitchell 1961, Nobel 1978).

6.3 Metabolic pathways

- Glycolysis: $\text{glucose} \rightarrow 2 \text{ pyruvate} + 2 \text{ ATP} + 2 \text{ NADH}$ (~ 10 steps)
- Citric acid cycle: $\text{pyruvate} \rightarrow \text{CO}_2 + \text{ATP} + \text{NADH} + \text{FADH}_2$ (~ 10 steps)
- Oxidative phosphorylation: $\text{NADH} + \text{O}_2 \rightarrow \text{H}_2\text{O} + \sim 30 \text{ ATP}$ (chemiosmosis)
- Photosynthesis (light $\rightarrow$ chemical energy): $\text{light} + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{glucose} + \text{O}_2$

Net glucose oxidation: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$, $\Delta G \approx -2870 \text{ kJ/mol} \rightarrow \sim 30\text{-}32 \text{ ATP/glucose}$.

6.4 Substrate view

Each metabolic step is a substrate K-type rearrangement, catalyzed by an enzyme that provides a substrate-K-type-optimal saddle.

The network architecture of metabolism (highly conserved across organisms) reflects substrate's natural pathway-architecture for biological chemistry.

6.5 Information $\leftrightarrow$ catalysis

Genes (Vol 13 Ch 2) encode enzymes (this chapter). Enzymes catalyze reactions (chemistry). Reactions reproduce DNA. Bootstrap loop: information $\leftrightarrow$ catalysis $\leftrightarrow$ reproduction.

BST view: this loop IS substrate's biological-scale K-type autopoiesis (self-creation). Closes the substrate computational reading of life: substrate K-types do biology.

6.6 K-audit anchors

- Vol 12 Ch 5 (kinetics, transition states)
- Vol 6 (thermodynamics)
- Vol 14 (information theory)

Level 3 — 5th-grader accessibility

Biochemistry = chemistry of life. Enzymes speed up reactions by $10^5\text{-}10^{17}\text{x}$. ATP: universal energy currency. Metabolism: $\text{glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{ATP}$ (~ 30

per glucose). In BST: each step is a substrate K-type rearrangement; enzymes provide optimal substrate paths. Information $\leftrightarrow$ catalysis $\leftrightarrow$ reproduction: substrate's biological autopoiesis loop.

What comes next

Chapter 7: Protein Structure (existing BST_Vol13 chapter).

Where to look this up

- Lehninger, *Principles of Biochemistry*
- BST: Vol 12 Ch 5; Vol 6; Vol 14